

LESSON PLAN

Academic Session: 2025-26

Week: _____

Developed by: FDC FAISAL

Taught By: _____

Subject: Maths	Class-IX	Duration: 40 mins
Unit & Title: Complex Numbers		Topic: Intro, Modulus, Conjugate & Operations
Lesson Objectives: TSWBAT By the end of the lesson, The Students Will Be Able To (TSWBAT): • Recognize complex numbers, their conjugate, and modulus. • Apply basic operations (addition, subtraction, multiplication) on complex numbers. • Demonstrate logical thinking while manipulating imaginary unit $i = \sqrt{-1}$.		
Skills Focused on: Problem solving, logical thinking, mathematical reasoning. • Problem solving • Logical thinking • Mathematical reasoning		
Resources: • Textbook & Calculator • Whiteboard & Markers		
Methodology:		
(a) Prior Knowledge (Recap, Brainstorming, Mind Mapping): • Teacher recalls previous knowledge by asking: – What happens when we try to find the square root of a negative number? – Can we solve the equation $x^2 + 1 = 0$ using real numbers? • Brainstorming on the need to expand our number system. • Mind mapping: Real Numbers $\rightarrow$ Imaginary Unit (i) $\rightarrow$ Complex Numbers ($a + ib$).		
(b) Explanation (Based on Cross-Curricular Links): • Introduction to the standard form of a complex number $z = a + ib$. • Explanation of conjugate ($\bar{z} = a - ib$) and modulus ($z = \sqrt{a^2 + b^2}$). • Step-by-step demonstration of adding and multiplying two complex numbers. • Cross-curricular links: – Physics/Electronics: Use of complex numbers to calculate alternating current (AC) impedance.		

(c) Activity:

- Students work individually to:
 - Find the conjugate and modulus of given complex numbers from the textbook.
 - Perform addition and multiplication on two given complex numbers.
- Teacher moves around the class, guiding and correcting algebraic techniques.

(d) Wrap-up:

- Oral questioning:
 - What is the value of i^2 ?
 - How does a conjugate change the sign of the imaginary part?
- Quick recap of key algebraic rules.

C.W: Perform basic addition and multiplication on complex numbers as given in textbook exercise.

H.W:

- Find the modulus and conjugate of a set of 5 complex numbers.

Differentiation:

- **Support:** Provide step-by-step templates for adding $(a + c) + i(b + d)$.
- **Challenge:** Ask advanced students to prove that $z \cdot \bar{z} = |z|^2$.
- **Visual learners:** Use color-coding on the board (e.g., Real parts in blue, Imaginary in red).

AFL:

- Observation during activity.
- Oral questioning to assess understanding.
- Checking accuracy of algebraic manipulation in classwork.

Critical Evaluation:

Date: _____ Signature: Subject Coordinator/HM _____

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Subject: Maths	Class-IX	Duration: 40 mins
Unit & Title: Complex Numbers	Topic: Laws, Identities, Inverses & Quotients	
Lesson Objectives: TSWBAT By the end of the lesson, The Students Will Be Able To (TSWBAT): • Define commutative, associative, and distributive laws for complex numbers. • Recognize additive/multiplicative identities and find inverses. • Find real and imaginary parts of expressions like $\frac{x_1+iy_1}{x_2+iy_2}$ and $(x + iy)^n$ for $n = \pm 1, \pm 2, \dots$		
Skills Focused on: Problem solving, logical thinking, mathematical reasoning. • Problem solving • Logical thinking • Mathematical reasoning		
Resources: • Textbook & Calculator • Whiteboard & Markers		
Methodology:		
(a) Prior Knowledge (Recap, Brainstorming, Mind Mapping): • Teacher recalls previous knowledge by asking: – What is the multiplicative identity for real numbers? – How do we rationalize a denominator with a square root (surd)? • Brainstorming: How does rationalization apply to imaginary numbers?		
(b) Explanation (Based on Cross-Curricular Links): • Discuss commutative, associative, and distributive laws extending from real to complex numbers. • Define Identity elements: $0 + 0i$ (Additive) and $1 + 0i$ (Multiplicative). • Demonstrate finding the multiplicative inverse using conjugates. • Explain how to separate real and imaginary parts by multiplying numerator and denominator by the conjugate of the denominator for $\frac{x_1+iy_1}{x_2+iy_2}$.		

(c) Activity: - Students work in pairs to: - Find the multiplicative inverse of $3 + 4i$. - Separate the real and imaginary parts of $\frac{2+i}{1-i}$. - Peer discussion encouraged to check rationalization steps.
(d) Wrap-up: - Oral questioning: Why do we multiply by the conjugate to perform division? - Emphasis on maintaining algebraic accuracy.
C.W: Find the multiplicative inverse and separate real/imaginary parts of given textbook problems.
H.W: Expand $(2 + 3i)^2$ and separate its real and imaginary parts.
Differentiation: Support: Provide the direct formula for multiplicative inverse $\left(\frac{a}{a^2+b^2} - i\frac{b}{a^2+b^2}\right)$. Challenge: Ask students to evaluate $(1 + i)^{-2}$.
AFL: - Checking accuracy of rationalizing the denominator. - Oral questioning.
Critical Evaluation:

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Subject: Maths	Class-IX	Duration: 40 mins
Unit & Title: Complex Numbers Interpretation		Topic: Equations & Geometric
Lesson Objectives: TSWBAT By the end of the lesson, The Students Will Be Able To (TSWBAT): • Find solutions of equations having complex values. • Apply the geometric interpretation of a complex number (Argand diagram). • Solve daily life problems involving complex numbers.		
Skills Focused on: Problem solving, logical thinking, mathematical reasoning. • Problem solving • Logical thinking • Mathematical reasoning		
Resources: • Textbook, Graph paper & Calculator • Whiteboard & Markers		
Methodology:		
(a) Prior Knowledge (Recap, Brainstorming, Mind Mapping): • Teacher recalls previous knowledge by asking: – How do we plot a point (x, y) on a Cartesian plane? – What is the quadratic formula?		
(b) Explanation (Based on Cross-Curricular Links): • Introduce the Argand Diagram: plotting $z = a + ib$ where the x-axis is real, y-axis is imaginary. • Demonstrate solving a quadratic equation with $D < 0$ (e.g., $x^2 + 4x + 5 = 0$), resulting in complex roots. • Cross-curricular links: – Art/Computer Science: Mention the Mandelbrot set and fractals plotted using complex numbers. – Engineering: Interpreting phase shifts in wave behavior (daily life/engineering application).		

(c) Activity: • Students work individually to: – Plot given complex numbers and their conjugates on an Argand diagram. – Solve a quadratic equation yielding complex roots. • Teacher moves around the class checking graph plots.
(d) Wrap-up: • Oral questioning: What geometric relationship exists between a complex number and its conjugate? (Reflection across the real axis).
C.W: Plot 3 complex numbers on graph paper; solve 2 complex-valued equations from textbook.
H.W: • Word problem applying complex numbers (as provided in the textbook's daily life section).
Differentiation: • Visual learners: Heavy reliance on graph plotting to bridge abstract math to visual representation. • Challenge: Ask students to geometrically show vector addition of two complex numbers.
AFL: • Checking geometric accuracy on graph paper. • Reviewing steps of the quadratic formula applied to negative discriminants.
Critical Evaluation:

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